

CMPU-395 Project Rough Draft

Group 12

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1. Introduction & Motivation

Traditionally, academic performance has always been viewed through internal student metrics such as attendance, grades, test scores, etc. Although these measures capture individual-level outcomes, they provide limited insight into the structural and environmental conditions influencing educational trajectories prior to classroom entry. Recent research increasingly examines the impact of external socioeconomic and geographic factors on academic success; however, scalable, non-invasive methods for capturing these factors at a granular level remain underdeveloped.

Recent scholarship has broadened the understanding of student performance determinants. Al-Ali et al. (2024) demonstrated that socioeconomic and psychological factors, including household income, parental education level, and access to resources, are strong predictors of academic outcomes, surpassing many traditional internal metrics in machine learning models. Although this study confirmed the significance of non-internal predictors, it relied on intensive self-reported survey data, which is inherently constrained in geographic scope, logistical scalability, and temporal frequency. Survey-based approaches are costly to administer at scale, frequently subject to response bias, and cannot be updated in real time as community conditions evolve.

Concurrently, computer vision research has shown that street-level imagery contains substantial socioeconomic information. Gebru et al. (2017) trained a deep learning model on millions of Google Street View images to estimate the demographic and socioeconomic composition of U.S. neighborhoods, using detected vehicle types as proxies for community wealth. Their "vehicular census" demonstrated that the physical built environment, parked vehicles, infrastructure condition, and the presence or absence of specific amenities provide an accurate signal of a neighborhood's socioeconomic status. This work established a foundational precedent: visual information passively observable from public streets can serve as a substitute for or complement to costly administrative and survey data.

This project integrates these two research domains into a unified, multimodal predictive framework. The central hypothesis is that if street-level imagery can map the socioeconomic characteristics of a neighborhood (Gebru et al., 2017), and if these socioeconomic factors are primary determinants of academic achievement (Al-Ali et al., 2024), then combining Computer Vision (CV) and Natural Language Processing (NLP) can yield a robust, non-invasive tool for predicting district-level educational outcomes across the United States. The primary research question is: To what extent can passively collected visual and textual data about neighborhoods predict district-level academic achievement, as measured by SEDA scores? It is anticipated that neighborhoods exhibiting visual indicators of higher socioeconomic status and more positive community discourse will be associated with higher district-level educational outcomes.

2. Motivation

Traditional research on academic outcomes relies on a well-established but increasingly limited set of tools, such as household demographics, student-level assessments, administrative records, and census statistics. While these data sources are valuable, they share a major limitation. They describe students and households in relative isolation, often overlooking the spatial and structural context in which schooling occurs. This study tackles this gap via treating the educational setting, neighborhood, built environment, and surrounding community as a primary unit of analysis. In doing so, it reframes educational inequality as both a geographic and an individual issue.

The rationale for this shift is grounded in both empirical evidence and real-world considerations. In numerous low-income communities across the United States, the lack of educational infrastructure outside of schools, such as libraries, tutoring centers, broadband internet access, and safe outdoor spaces, creates what researchers have termed "opportunity deserts". Opportunity deserts are areas in which environmental conditions systematically constrain children's developmental potential before formal schooling begins (Chetty et al., 2014). These structural deficits are not captured by student-level metrics. While a child's test score reflects the outcome, the neighborhood reflects the underlying conditions.

Census data has traditionally served as the primary tool for tracking geographic disparities in resource access. However, it faces two well-documented limitations. First, the decennial cycle results in rapidly changing communities, such as those experiencing gentrification, deindustrialization, or population shifts, going up to a decade without updated representation. Second, census data collection is subject to undercount errors and various privacy adjustments, which can distort estimates in precisely those communities where valid measurement is most critical. Alternative administrative data sources, such as the American Community Survey, offer more frequent updates but still suffer from significant lag and sampling error at fine geographic scales.

This project delivers a supplementary approach by leveraging continuously updated, publicly available data streams, specifically Google Street View imagery and community-generated online text, to produce near-real-time assessments of neighborhood conditions relevant to educational opportunity. Street View imagery, updated on rolling cycles by Google, captures the visual state of neighborhoods with a level of granularity and frequency unmatched by government data sources. Online social discourse, as observed on platforms such as Reddit, gives insights into community members' lived experiences and perceptions of local resources, safety, and quality of life. Combining these data sources offers a high-resolution, current map of where educational investment is most and least prevalent.

In addition to its forecasting objectives, this framework functions as a scalable method for identifying districts where ecological indicators indicate that students face structural disadvantages not yet addressed by policy responses. This tool could promote targeted resource assignment, early intervention, and evidence-based advocacy, offering education researchers, policymakers, and community organizations a dynamic, low-cost alternative to traditional administrative data.

3. Data Description

3.1 School District / Streetview Dataset:

For the computer vision component of the project, we constructed a district-level dataset linking school district academic outcomes with Google Street View imagery. The unit of analysis is the **school district**, not the individual image. This distinction is important because the educational outcome labels come from the Stanford Education Data Archive (SEDA), which reports achievement estimates at the district level, while the image data consists of multiple sampled street-level views within each district.

The main educational outcome variable used in the computer vision analysis is a SEDA achievement score, treated as a continuous measure of district-level educational performance. We filtered the SEDA data to use a consistent year and removed observations with missing or unreliable outcome values. We also retained district identifiers, district names, state abbreviations, and reliability-related variables when needed for cleaning and documentation. Variables unrelated to the modeling task, such as redundant geographic codes, duplicate measurement variants, and unused administrative fields, were removed to keep the modeling dataset focused and interpretable.

To connect SEDA outcomes with visual neighborhood information, we used school district boundary shapefiles from the National Center for Education Statistics. These shapefiles define each district as a polygon. For each selected district, we sampled geographic points within the district boundary and queried the Google Street View metadata endpoint before downloading images. This was necessary because Street View requests may return the nearest available panorama rather than an image exactly at the requested coordinate. Therefore, we retained only images whose returned panorama coordinates fell inside the correct district polygon.

Since our progress report, we expanded the computer vision dataset substantially. The earlier version of the pipeline produced only about 90 usable districts after filtering. For the rough draft analysis, we loosened the minimum image threshold and expanded the number of attempted districts. The current CNN dataset includes **250 districts** and **7,033 street-view images** after cleaning. Each included district has at least **15 images**, with an average of approximately **28 images per district**. Image counts range from 15 to 40 per district. This expansion was important because the effective sample size for supervised learning is the number of districts, not the number of images, since each district has one outcome label.

The final cleaned dataset was split at the district level into training, validation, and test sets. This prevents leakage from having images from the same district appear in both training and testing. The final split contains:

Split	Districts	Images
Train	175	4945
Validation	37	1040
Test	38	1048

This split structure ensures that model performance reflects generalization to unseen districts rather than memorization of visual patterns from districts already present in training.

3.2 Reddit Sentiment Dataset

We constructed a district-level natural language dataset drawn from Reddit, a large-scale online platform where users discuss local community issues, schools, and neighborhood conditions. The goal of this component was to capture community-level discourse as a proxy for social perception of neighborhood quality. This is a dimension of educational context not directly observable from street imagery alone.

We constructed search queries using the district name and state abbreviation, running 3 queries per district to maximize the number of relevant posts retrieved. We collected posts using Reddit's public JSON search endpoint because our Reddit API access was not approved in time for this analysis. Using Reddit's public JSON search endpoint does not require authentication, but it is subject to rate limits and doesn't support filtering by subreddit or date range. This constrained the volume and the specificity of posts we retrieved per district.

The retrieved posts were cleaned to remove deleted content, moderator messages, and non-English text. Only districts with at least three retrievable posts were retained for correlation analysis, yielding a final sample of 289 districts. We then used a RoBERTa-based sentiment model for each post, producing a sentiment score ranging from -1 (most negative) to +1 (most positive), weighted by model confidence. Scores were aggregated to a per-district mean sentiment value.

The resulting dataset contains one row per district with the following key variables: district identifier (sedalea), district name, state abbreviation, mean sentiment score, post count, standard deviation of sentiment, and the corresponding SEDA achievement score. The mean sentiment across all districts was -0.22 (SD = 0.13), reflecting the generally critical tone of Reddit discourse around schools. Reddit users are more likely to post complaints than praise. The negative skew is an important characteristic of the data that complicates the direct interpretation of sentiment as a proxy for community quality.

Coverage was uneven across district types. Urban and suburban districts with larger populations generated more Reddit posts and, hence, more reliable sentiment estimates. Smaller, rural, and less-connected districts returned fewer posts, and some returned none at all. This sampling bias reflects a known limitation of using social media data as a proxy for geography. Reddit's user base is not representative of the U.S. population, and is skewed toward younger, more urban, and higher-income demographics (Pew Research Center, 2021). As a result, sentiment estimates for rural and underrepresented districts should be interpreted with caution.

4. Methods

4.1 Computer Vision Pipeline:

The goal of the image analysis is to test whether visual features of the built environment contain predictive information about district-level academic outcomes. Our initial approach used a pretrained ResNet-50 model as a feature extractor. Each image was passed through ResNet-50 with the final classification layer removed, producing a 2048-dimensional embedding. These image embeddings were

then averaged within each district to form a single district-level feature vector. We used this representation with ridge regression and random forest regression as preliminary baselines.

However, this approach did not fully use the image modeling capacity of convolutional neural networks. In response to feedback, we implemented a more direct CNN-based approach using a **multiple-instance learning (MIL)** framework. This better matches the structure of our data: each district can be understood as a “bag” of street-view images, and each bag has one district-level educational outcome label.

In the MIL CNN model, each training example is a district rather than an individual image. For each district, the model samples a fixed number of images and passes them through a ResNet-50 backbone pretrained on ImageNet. The CNN extracts visual features from each image. These image-level features are then pooled across images to create a district-level representation. In the current implementation, we use mean pooling. A regression head then predicts the district’s SEDA outcome score from this pooled representation.

This method has two advantages over treating images independently. First, it respects the fact that the true label belongs to the district, not to each image. Second, it allows the model to use multiple visual observations from a district rather than relying on a single image. We evaluated the model using district-level metrics: mean absolute error (MAE), root mean squared error (RMSE), R^2 , and Pearson correlation between predicted and observed SEDA scores.

We also generated several visualizations to evaluate the results, including prediction-versus-true scatter plots, residual plots, PCA visualizations of district-level representations, and training curves. These plots help us determine whether the model learns meaningful signal, whether predictions are biased, and whether performance changes across training epochs.

4.2 NLP Pipeline: Reddit Sentiment Analysis

The goal of the NLP component is to capture community-level discourse about local schools and neighborhoods as an additional signal to the visual features extracted from street-level imagery. For each of the 289 districts in our sample, we collected posts from Reddit using its public JSON search endpoint, which does not require API authentication. Though functional, this approach is rate-limited and does not support filtering by subreddit or date range, which constrained the volume and specificity of posts retrievable per district.

To maximize coverage, we constructed three query variants per district using the district name (with common administrative suffixes such as "unified school district" and "elementary" removed) and state abbreviation. Example queries for a given district took the form: "*{district name} {state} schools education*", "*{district name} {state} school district*", and "*{district name} {state}*". Duplicate posts were identified by title and removed. Posts with fewer than 20 characters of combined title and body text were excluded, as were deleted and removed posts. To avoid overloading Reddit's servers, a 1.5-second delay was inserted between queries and a 2-second delay between districts.

Retrieved post titles and body text were concatenated into a single string per post and passed through a RoBERTa-based sentiment classification model fine-tuned on social media text (*cardiffnlp/twitter-roberta-base-sentiment-latest*). This model classifies each post as positive, neutral, or

negative, and outputs a confidence score. To produce a continuous sentiment score, we mapped the predicted label to a base value (+1 for positive, 0 for neutral, -1 for negative) and multiplied it by the model's confidence, yielding a score in the range [-1, 1]. Posts shorter than 10 characters were excluded from scoring. Scores were then averaged across all scoreable posts within a district to produce a single mean sentiment value per district. Only districts with at least three scoreable posts were retained for further analysis. We got a final sample of 289 districts.

4.3 Multimodal Fusion

To evaluate whether combining visual and textual signals increases predictive performance relative to either modality alone, we used a multimodal regression model that concatenates district-level CNN embeddings with per-district RoBERTa sentiment scores. The 2048-dimensional mean-pooled ResNet-50 embeddings (described in Section 3.1) were merged with the sentiment dataset on district identifier, yielding a matched sample of 132 districts present in both datasets. The combined feature vector for each district has 2049 dimensions: 2048 visual embedding dimensions and one sentiment score.

We evaluated three model configurations using RidgeCV regression with cross-validated alpha selection over the range [0.01, 10,000]: a CNN-only model using the 2048-dimensional embeddings alone, an NLP-only model using mean sentiment as the sole predictor, and a multimodal model using the full concatenated feature vector. All features were standardized using a StandardScaler fit on the training set. Districts were split into training (70%), validation (15%), and test (15%) sets, using a fixed random seed to preserve reproducibility. Model performance was measured using MAE, RMSE, R^2 , and Pearson correlation between predicted and observed SEDA scores.

5. Results

5.1 CNN-Based District Prediction:

After expanding the dataset, we evaluated two types of computer vision prediction pipelines. The first approach used a pretrained ResNet-50 feature extractor, where each image was converted into a 2048-dimensional embedding and embeddings were averaged within each district. These district-level visual representations were then used as input to traditional regression models. The second approach used a more direct multiple-instance learning (MIL) CNN, where a ResNet-50 backbone processed multiple images from each district and produced a district-level prediction through feature pooling and a regression head.

Pretrained ResNet-50 Embedding Baselines:

The pretrained ResNet-50 embedding approach produced the strongest results overall. Both ridge regression and random forest regression achieved similar performance on the held-out test set:

Model	MAE	RMSE	R^2	Pearson
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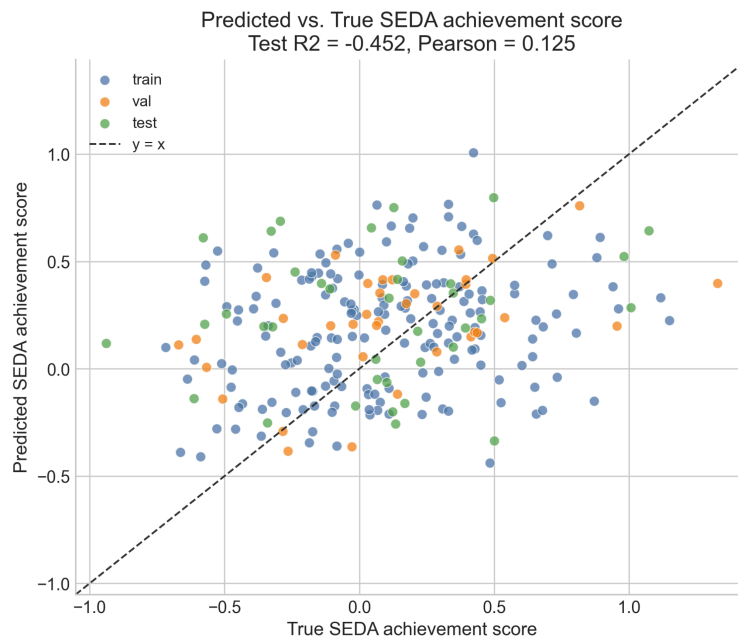
ResNet 50 + Ridge	0.3109	0.3898	0.2022	0.5037
ResNet50 + Random Forests	0.3114	0.3881	0.2094	0.5037

These results suggest that pretrained visual embeddings contain meaningful information about district-level educational outcomes. An R^2 of approximately 0.20 indicates that the image-based features explain about one-fifth of the variation in the held-out district scores, while a Pearson correlation of about 0.50 indicates a moderate positive relationship between predicted and observed outcomes.

MIL CNN Results

In response to feedback, we also implemented a CNN-based multiple-instance learning model. This model treats each district as a bag of Street View images and learns a district-level representation by pooling CNN features across images. The results were:

Split	MAE	RMSE	R^2	Pearson
Train	0.355	0.434	-0.256	0.218
Validation	0.306	0.392	0.127	0.453
Test	0.435	0.533	-0.452	0.125



The validation results initially appeared moderately promising. On the validation set, the model achieved a positive R^2 of **0.127** and a Pearson correlation of **0.453**, suggesting that the CNN was able to capture some relationship between street-view imagery and educational outcomes. The best validation MAE occurred around epoch 7, with validation MAE approximately **0.300**. The best validation RMSE and R^2 occurred around epoch 5, and the best validation Pearson correlation occurred around epoch 10.

However, the test set results were substantially weaker. On held-out test districts, the model achieved an MAE of **0.435**, RMSE of **0.533**, R^2 of **-0.452**, and Pearson correlation of only **0.125**. A negative R^2 means that the model performed worse than a simple baseline that predicts the mean outcome. This suggests that the current CNN model does not yet generalize reliably to unseen districts.

The test predictions also showed upward bias. The mean true test outcome was **0.073**, while the mean predicted value was **0.244**. This indicates that the model tended to overpredict educational scores for the test districts. In practical terms, the model was not only inaccurate on the test set but also systematically shifted upward.

Overall, the CNN experiment produced a mixed result. The validation performance suggests that visual information may contain useful predictive signal, but the poor test performance indicates that the current CNN model is unstable and likely sensitive to the particular district split.

Model Comparison

The main finding from the computer vision analysis is that the simpler pretrained ResNet-50 embedding pipeline currently outperforms the trainable MIL CNN. Although the CNN model is more flexible, it likely requires more district-level training examples to learn a stable relationship between visual features and educational outcomes. In contrast, the pretrained ResNet-50 pipeline benefits from pretrained ImageNet features and only trains a relatively simple regression model on top of aggregated district representations.

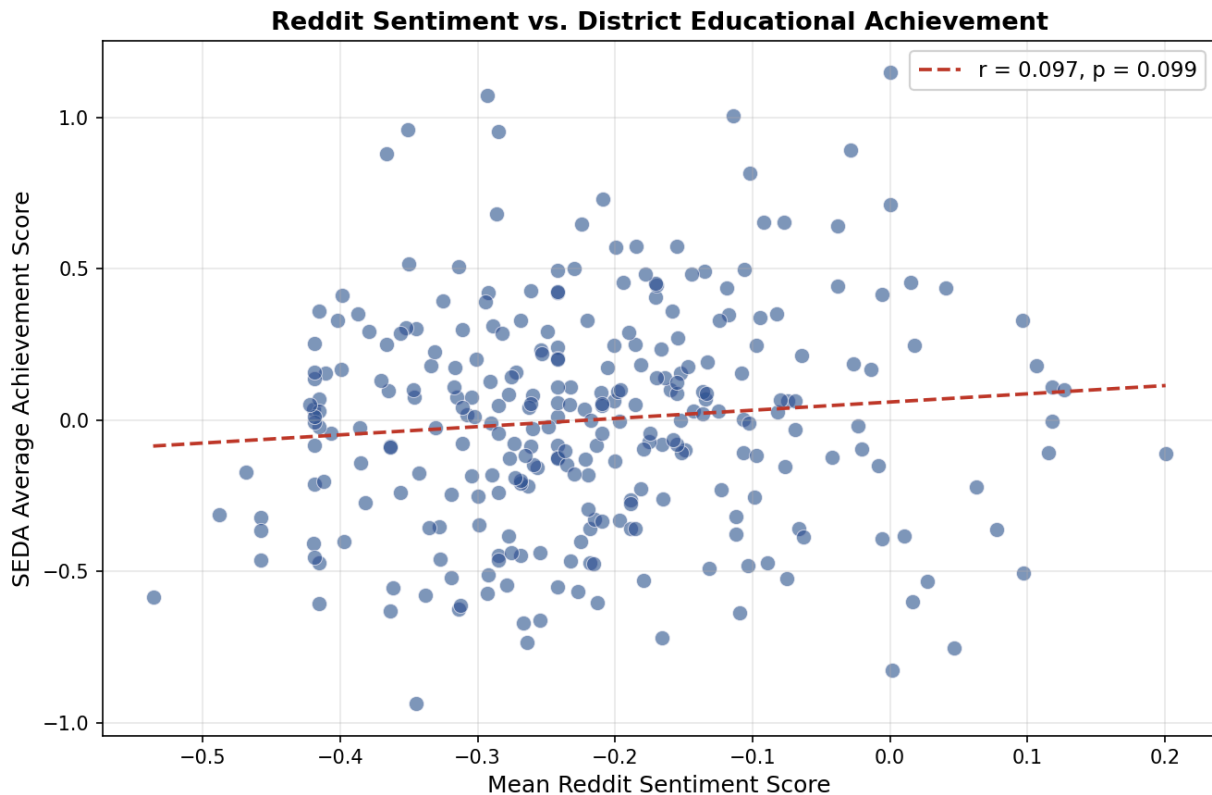
This result is important because it shows that using a CNN does not automatically improve performance. Given the current dataset size, transfer learning through fixed embeddings appears to be more stable and reliable than fine-tuning a CNN end-to-end.

5.2 NLP Sentiment Results

To evaluate the predictive value of Reddit discourse, we first looked at the correlation between district-level mean sentiment scores and SEDA achievement scores across the 289 districts with sufficient post coverage. The Pearson correlation between mean sentiment and SEDA achievement was $r = 0.097$ ($p = 0.099$), and the Spearman correlation was $\rho = 0.104$ ($p = 0.078$). Neither result reached statistical significance at $\alpha = 0.05$, suggesting that Reddit sentiment alone does not dependably track district-level academic outcomes at this sample size.

When used as a sole predictor in a ridge regression framework on the 132-district matched sample, the NLP-only model performed poorly across all splits:

Split	MAE	RMSE	R ²	Pearson
Train	0.318	0.400	0.000	0.150
Validation	0.403	0.457	-0.516	0.167
Test	0.395	0.487	-0.008	0.268



The R^2 values on both the training and test sets (both near 0) indicate that a single mean sentiment score provides little linear predictive power for SEDA outcomes in this model. The negative validation R^2 further confirms that sentiment-based predictions are worse than simply predicting the mean outcome for unseen districts. The test-set Pearson correlation of 0.268, while slightly higher than on the training and validation sets, is not nearly sufficient to indicate reliable generalization and is most likely noise given the small test set size of 21 districts.

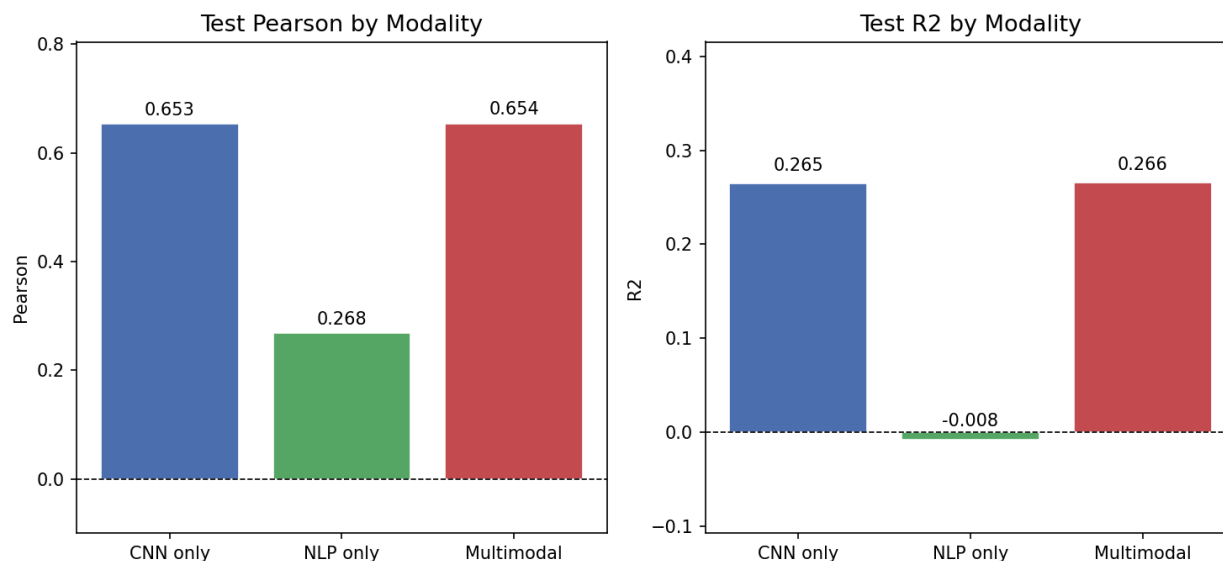
These results are consistent with the weak correlation observed in the full 289-district sample. The overall negative mean sentiment (-0.22) across districts likely reflects the complaint-driven nature of Reddit discourse rather than genuine variation in these school district community conditions. Because Reddit users tend to post when they have complaints rather than praise, the sentiment signal may be compressed and noisy as a proxy for educational quality.

5.3 Multimodal Fusion Results

To evaluate whether combining visual embeddings with Reddit sentiment improves predictive performance, we concatenated the 2048-dimensional district-level ResNet-50 embeddings with per-district mean sentiment scores and re-ran RidgeCV regression on the 132-district matched sample. The results across all three model configurations are summarized below:

Model	Split	MAE	RMSE	R ²	Pearson
CNN only	Train	0.229	0.291	0.471	0.856
CNN only	Validation	0.355	0.417	-0.259	0.426
CNN only	Test	0.337	0.416	0.265	0.654
NLP only	Train	0.318	0.400	0.000	0.150
NLP only	Validation	0.403	0.457	-0.516	0.167
NLP only	Test	0.395	0.487	-0.008	0.268
Multimodal	Train	0.229	0.291	0.471	0.856
Multimodal	Validation	0.354	0.416	-0.257	0.427
Multimodal	Test	0.337	0.416	0.266	0.654

Unimodal vs. Multimodal — Test Set Performance



To evaluate whether combining visual embeddings with Reddit sentiment improves predictive performance, we concatenated the 2048-dimensional district-level ResNet-50 embeddings with per-district mean sentiment scores and re-ran RidgeCV regression on the 132-district matched sample. The results across all three model configurations are summarized above.

The multimodal model produced results nearly identical to the CNN-only model across all splits and metrics. On the test set, adding sentiment to the CNN embeddings changed R^2 by only 0.001 and Pearson by 0.001, differences that are negligible in practical terms. The CNN-only model on this 132-district matched sample achieved a test-set Pearson of 0.654 and R^2 of 0.265 — a substantially stronger result than the MIL CNN reported in Section 5.1 (Pearson = 0.125, R^2 = -0.452). This improvement reflects both the larger district sample available for the embedding pipeline and the stability advantage of fixed pretrained representations over end-to-end fine-tuning at limited label counts. The sentiment score adds no marginal predictive value beyond what is already encoded in the 2048-dimensional visual embedding vector.

6. Interpretation & Non-Results

6.1 School District & Streetview

The most important result from the computer vision analysis is that expanding the dataset alone did not automatically solve the prediction problem. Even though we increased the dataset to **250 districts** and more than **7,000 images**, the CNN model still struggled to generalize to unseen districts. This reinforces an important methodological point: although the dataset contains thousands of images, the number of independent supervised labels is still only the number of districts. Since each district has one SEDA outcome score, the effective supervised sample size is closer to **250 observations**, not 7,000 images.

The weak test performance of the MIL CNN may be explained by several factors. First, district-level achievement scores are noisy relative to street-level images. A district's academic outcome is influenced by many variables that are not directly visible in Street View imagery, including school funding formulas, curriculum, teacher retention, household education, local policy, and student demographics. Second, districts are geographically and visually heterogeneous. A small sample of 15–40 Street View images may not represent the full range of conditions within a district, especially in large or internally diverse districts. Third, the CNN may have overfit to visual patterns in the training and validation districts that did not transfer well to the held-out test districts.

The CNN results also differed from the simpler pretrained ResNet-50 embedding baseline. In the earlier baseline, pretrained ResNet embeddings paired with ridge regression or random forest regression produced moderate performance, with test-set Pearson correlation around **0.50** and R^2 around **0.20**. This suggests that pretrained visual embeddings do contain meaningful signal about district-level outcomes. However, the MIL CNN did not outperform this simpler approach on the held-out test set. This is a useful non-result: a more complex image model is not automatically better when the number of district-level labels is limited. In our current setting, fixed pretrained representations appear more stable than fine-tuning a CNN.

6.2 NLP & Multimodal Interpretation

The NLP results constitute a meaningful non-result. Reddit sentiment, as operationalized here, does not appear to be a dependable predictor of district-level academic achievement. A number of factors likely explain this. First, Reddit's user base is skewed toward younger, urban, and higher-income populations, meaning that discourse about schools in lower-income or rural districts is systematically underreported or absent. Second, the public JSON scraping approach restricted the volume and specificity of posts retrievable per district, making mean sentiment estimates for many districts highly sensitive to individual posts. Third, the nature of Reddit discourse around schools tends toward complaint regardless of actual district quality, compressing the range of sentiment scores and reducing their discriminative power.

The failure of the multimodal fusion model to outperform the CNN-only model is similarly informative. When one modality, the 2048-dimensional CNN embeddings, is dense and informative, and another, the single sentiment score, is low-dimensional and noisy, ridge regression will effectively assign near-zero weight to the weaker modality. This is not a failure of the multimodal hypothesis in general, but an outcome of the extreme asymmetry in feature scale and information quality between the two modalities. A more balanced fusion strategy, like training a separate district-level text embedding at comparable dimensionality, or a late-fusion ensemble that weights modalities independently, might produce different results. The current finding rules out only that this particular operationalization of sentiment adds significantly, not that richer NLP features could do so.

7. Discussion & Conclusion

7.1 Discussion

This project asked whether passively collected neighborhood data, especially Google Street View imagery

and Reddit discourse, can predict district-level academic achievement. Overall, our results suggest that these data sources contain some signal related to educational outcomes, but that the signal is uneven, noisy, and highly dependent on modeling choices.

The strongest result from the computer vision analysis came from the pretrained ResNet-50 embedding approach. When we used ResNet-50 as a fixed feature extractor and aggregated image embeddings at the district level, both ridge regression and random forest regression achieved moderate predictive performance, with test-set Pearson correlations around **0.50** and R^2 values around **0.20**. This suggests that street-level imagery does capture some features of the built environment that are associated with district-level academic achievement. This result is consistent with the broader logic of Gebru et al. (2017), who showed that visual information from Google Street View can serve as a proxy for neighborhood socioeconomic characteristics. In our case, visual features alone do not fully explain educational outcomes, but they appear to encode some information related to neighborhood context.

However, the more complex MIL CNN model did not outperform the simpler pretrained embedding approach. Although the validation results appeared promising, the test results were weak, with negative R^2 and low Pearson correlation. This suggests that the CNN did not generalize reliably to unseen districts. One important lesson from this non-result is that using a more complex image model does not automatically improve prediction. Even though the dataset contains more than **7,000 images**, the number of independent labels is only **250 districts**, because each district has one SEDA score. This means the project is not a standard image classification task. It is closer to a weakly supervised, district-level prediction problem where many images are used to represent one labeled unit. Under this setup, frozen pretrained features may be more stable than fine-tuning a CNN with limited district-level labels.

The NLP results were also limited. Reddit sentiment alone had weak correlation with SEDA achievement and performed poorly as a standalone predictor. This is not surprising because Reddit discourse is not a neutral or representative measurement of community conditions. The mean sentiment score was negative overall, likely because people are more likely to post complaints about schools and neighborhoods than positive evaluations. The Reddit dataset also has strong coverage bias: urban and suburban districts tend to have more posts, while rural and smaller districts often have little or no usable discourse. As a result, Reddit sentiment should not be interpreted as a direct measure of educational quality.

The CNN-only ridge regression model on the 132-district matched sample achieved the strongest test-set result overall, with a Pearson correlation of 0.654 and R^2 of 0.265. This represents a substantial improvement over the MIL CNN results reported earlier, and is best explained by two factors: the larger district sample available for the embedding pipeline and the stability of frozen pretrained representations relative to end-to-end fine-tuning under limited supervision. The multimodal model produced virtually identical results, with test Pearson of 0.654 and R^2 of 0.266, a difference of 0.001 on both metrics. This makes clear that the Reddit sentiment score added no meaningful predictive power on top of the CNN embeddings, and that the strength of this result should be attributed to the visual modality alone rather than to multimodal fusion. The apparent success of the multimodal framework in this case is therefore better understood as a confirmation that street-level visual features carry real predictive signal, not as evidence that combining modalities improves prediction.

Taken together, the results partially support our original hypothesis. Street View images and Reddit

discourse do not provide a complete or reliable substitute for administrative education data, but they do contain partial signals that are associated with district-level achievement. The project therefore strengthens the idea that neighborhood context matters, while also showing the methodological difficulty of measuring that context with proxy data.

7.2 Reliability, Validity & Limitations

There are several important limitations in the current analysis. First, the project uses district-level SEDA achievement scores as the target variable. These scores are useful because they provide standardized comparisons across districts, but they are still aggregate estimates. They do not measure individual student opportunity, school-level variation within districts, or causal mechanisms. A district with a high achievement score may still contain severe inequalities across schools or neighborhoods, while a lower-scoring district may have strong support systems not captured by the aggregate score.

Second, the Street View image data may not fully represent each district. Each district is represented by 15–40 images, but districts vary greatly in physical size, population density, and internal heterogeneity. In a large or diverse district, a small set of sampled images may capture only part of the built environment. Google Street View coverage is also uneven; urban and suburban areas are often better represented than rural areas. Therefore, the visual dataset may systematically underrepresent precisely the kinds of places where educational access is most difficult to measure.

Third, the Reddit dataset has substantial sampling bias. Reddit users are not representative of the general population, and local Reddit activity is much higher in some communities than others. Districts with more online discussion are not necessarily districts with more educational issues; they may simply have more active internet users. The sentiment model also reduces complex discourse to a single mean score, which may erase important differences between posts about safety, taxes, school quality, transportation, or general local politics.

Fourth, there are modeling limitations. The number of district-level labels remains small relative to the complexity of the models and feature spaces. The image embeddings have 2048 dimensions, while the multimodal matched dataset has only 132 districts. This raises the risk of overfitting and unstable performance estimates. The CNN results especially show that model complexity can exceed the amount of supervision available. For this reason, our results should be interpreted as exploratory correlations rather than robust predictive claims.

These limitations restrict the inferential conclusions we can make. We cannot claim that the visual appearance of a neighborhood causes academic achievement, nor can we claim that Reddit sentiment accurately measures community educational opportunity. The most we can conclude is that, in our sampled data, certain visual and textual proxy signals are associated with district-level SEDA outcomes. The models may not generalize to new districts, different years, or regions with different Street View and Reddit coverage. A stronger analysis would require a larger and more representative district sample, more systematic image sampling, additional socioeconomic controls, and repeated validation across multiple random splits.

7.3 Ethical & Societal Considerations

This project has important ethical implications. The potential benefit of this work is that scalable data sources such as street imagery and online discourse could help identify places where educational resources are lacking. If used carefully, this type of model could supplement slower administrative datasets and help researchers or policymakers prioritize areas for further investigation.

However, there are also risks. Predicting educational opportunity from neighborhood appearance can easily reinforce stereotypes about communities. A model might learn visual patterns associated with poverty or disinvestment and then reproduce those associations without understanding the historical or policy causes behind them. If used carelessly, such a model could label neighborhoods as “low opportunity” in ways that stigmatize residents rather than support them. Similarly, Reddit discourse may overrepresent more privileged or more vocal users, meaning that communities with less online presence may become invisible in the analysis.

For these reasons, the models in this project should not be used to make high-stakes decisions about schools, families, or communities. They should be treated as exploratory tools that can suggest where further investigation may be needed, not as definitive assessments of educational quality. Any practical use would require transparency, community input, careful validation, and safeguards against reinforcing existing inequalities.

7.4 Conclusion

In conclusion, this project shows that neighborhood-scale visual and textual data can provide partial insight into district-level educational outcomes. The pretrained ResNet-50 embedding models suggest that Street View imagery contains meaningful socioeconomic and educational signal, while the Reddit sentiment results show that online discourse alone is too noisy to serve as a reliable predictor. The multimodal model offers the most promising preliminary result, but its instability across splits means that further validation is necessary.

For the next stage of the project, we plan to improve both the data and modeling pipeline. First, we will expand and balance the district sample where possible, especially for rural and underrepresented districts. Second, we will improve image sampling so that images better represent the full district rather than isolated points. Third, we will test more robust modeling strategies, including repeated train/test splits, stronger regularization, attention-based pooling for image bags, and multimodal feature selection. Fourth, we will improve the Reddit pipeline using approved API access so that posts can be collected with better filtering by subreddit, date, and relevance.

The main takeaway is that educational opportunity cannot be fully inferred from any single proxy. Street View imagery captures the physical environment, Reddit captures partial community discourse, and SEDA captures measured academic outcomes. Each source is incomplete on its own. A careful multimodal approach may provide a richer understanding of educational inequality, but only if its limitations, biases, and ethical risks are taken seriously.

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